FinRisk AI: Unified NLP Risk Engine & Portfolio Stress-Testing Platform

S&P Global & CRISIL Campus Hackathon 2026 | Candidate: Rudra Singla | Techno India NJR Institute of Technology

- Track: AI/NLP Risk Engine with Strategic Portfolio Stress Testing (Module B)
- Transforms unstructured market feeds into quantitative credit risk and regulatory capital triggers.
- Combines Hugging Face FinBERT domain sentiment with Basel III Expected Credit Loss (ECL) mechanics.

Problem Statement & The Latency Gap in Risk Monitoring

Why Traditional Credit Reviews Lag Behind Real-World Market Distress

• Unstructured Data Lag: Critical credit events appear in breaking news and social feeds 48-72 hours before rating agency adjustments.

• Information Asymmetry: Banks manage large wholesale portfolios without real-time linkage between qualitative news and portfolio capital adequacy.

• Our Solution: An automated, sub-second NLP Risk Engine mapping directly to wholesale asset repricing and capital adequacy (CRAR) buffers.

System Architecture & High-Performance Data Flow

End-to-End Pipeline: Ingestion -> NLP Engine -> Structured JSON -> Portfolio Stress Engine

- Data Ingestion: Dual-channel streaming ingesting financial news and micro-blog feeds into a normalized schema.
- Risk NLP Engine: FinBERT sentiment scoring (S in [-1, 1]), zero-shot event taxonomy classification, and explainable impact scoring.
- Stress-Testing Engine: Automated shock propagation executing Basel III ECL on loans and Modified Duration repricing on debt.
- Consumption Layer: Interactive Streamlit analytics dashboard and production-grade FastAPI REST endpoints.

AI/NLP Methodology & Impact Scoring Mechanics

Explainable Quantitative Scoring Replacing Black-Box Heuristics

• Sentiment Analysis: FinBERT model outputs calibrated score S = P(pos) - P(neg), delivering domain-accurate scores (e.g. -0.954).

• Event Categorization: Predefined taxonomy spanning Credit Events, Macroeconomic rate shifts, Geopolitical sanctions, and Earnings.

• Explainable Impact Score (1-10): I = Clip[1,10](0.50 * BaseSeverity + 0.30 * (|S| * 10) + 0.20 * SourceWeight).

• Attribution & Evidence: Engine automatically extracts driver tokens (e.g. 'debt default', 'spread widening') for risk committee auditability.

Wholesale Banking Portfolio & Basel III Stress Simulation

Institutional-Grade Risk Modeling Over Arbitrary Percentage Cuts

- Synthetic Portfolio Exposure: ₹100 Lakhs distributed across Corporate Loans, G-Secs, Corporate Bonds, Equities, and Swaps.
- Basel III Credit Impairment: dLoss = EAD * dPD * LGD (applied at 45% standard institutional LGD).
- Modified Duration Repricing: dP = -D_mod * dy * P, accurately modeling bond price erosion under interest rate hikes.
- Regulatory Capital Impact: Evaluates Tier-1 Capital depletion and automatically flags breaches against the 11.5% minimum CRAR requirement.

Prototype Demonstration & Key Evaluation Results

Empirical Verification Across Diverse Market Scenarios

• Credit Event Shock: Headline regarding commercial defaults triggered 9/10 impact, reducing portfolio value from ₹100L to ₹90.59L (-9.4%).

• Capital Buffer Alert: Stressed CRAR declined from 17.65% to 6.35%, automatically triggering an institutional regulatory capital alert.

• Macroeconomic Rate Shock: 75 bps surprise rate hike dynamically repriced 10Y Sovereign G-Secs based on duration (7.2y) without touching unrelated credit loans.

• False-Positive Resilience: Product launches and positive earnings retain 'NOMINAL' status without executing capital stress.

Domain Impact, Governance & Production Roadmap

Scaling from Prototype to Enterprise Risk Management (ERM)

• Business Value: Provides chief risk officers (CROs) with dynamic, intraday scenario stress-testing rather than static quarterly reviews.

• Enterprise Integration: Built-in FastAPI endpoints ready for streaming enterprise Kafka message queues.

• Governance & Regulatory Compliance: Full audit trail of extracted tokens, confidence scores, and formula breakdowns ready for regulatory examination.

• Future Roadmap: Multi-entity relationship graphs, Basel IV output floors, and real-time counterparty exposure limits.